Name : Dimpho Sekhosana

Student: 2442463

FACTORS THAT THE NICHE BREADTH AND OVERLAP IN BIRD COMMUNITIES

There are approximately 856 pence pieces in South Africa, and WITS has about 198 of the species meaning that WITS has 23% of this bird species. The bird species found in wits are the Indian myna, Egyptian goose, Rockdove, scared ibis cattle egret and Hadeda ibis just to name a few. Niche breath refers to a diversity of natural resources used and needed or the environment conditions the tolerated by an individual, species or a population. With niche breadth we investigate how it shapes geographic distribution and richness of species from local to landscape scales, but I will be focusing on its effect on bird communities.

Birds are a group of warm-blooded vertebrates that characterised by their feathers covering their bodies, toothless beaks jaw that in all sorts of different shapes and sizes, how they lay hard-shelled eggs, their high metabolic rate and how they have a four chambered heart with a very strong but lightweight skeleton. According to studies there is around 10,000 known bird species around the globe, because of the number of bird species being high. Bird species are considered to be one of the most abundant animal species on the planet, and the worlds most abundant bird being the familiar house sparrow with a population of 1,6 billion in the world, these bird communities are affected by niche breadth and niche overlap.

Niche overlap is the degree to which species share the factors such as resources which regulate their population growth. These factors are mainly influence by competition among species of the same (intraspecific competition) and competition among different species

(interspecific competition). The competitive differences between species is how species are efficient in converting resources to population growth in their average environment. Measure of niche breadth and overlap depends on the distribution of resources among individual bird communities but ids independent on the relative abundance of species and the number of resources states considered.

Niche of bird communities is affected by both biotic and abiotic factors, the biotic factors are predators (animals the hunt and kill other animals for food), competitors (both interspecific and intraspecific competition among bird species), parasites, commensals. The abiotic factors being temperature, climate, and soil type. The biotic factors are all influenced the availability of essential resources in a bird community, when the resources are low niche breadth is low and niche overlap is also low. With the demand of resources such as worms, water and shelter increase bird species are less likely to share resources and competition increase. Abiotic factors such as climate, in seasons such as winter trees dry out, lakes are frozen and the is less food for birds. Birds tend to harvest resource during summer to prepare for the upcoming winter season meaning competition for resource is high with bird species unlikely to share resource as each bird is trying to ensure their survival throughout winter.

With bird species being so abundant in the globe, the fittest and most efficient at obtain resources is more like to survive and grow its population. As bird species are more likely the same, the use the same resources and have the same dietary needs, its more likely that their niches are low.

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21 MAERCH 2022

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